



Large Language Models for Elderly Care in Smart Home Environments: A Systematic Review

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Abstract. Large Language Models (LLMs) are increasingly explored as a means to enhance smart home systems for elderly care, offering more natural interaction, contextual reasoning, and adaptive support compared to traditional rule-based approaches. Although prior work across artificial intelligence, humancomputer interaction, and ambient assisted living has highlighted their potential, existing research remains scattered and lacks an integrated overview. To address this gap, we conducted a systematic review following PRISMA 2020 guidelines, examining a wide body of literature from January 2020 to October 2025, from which 17 featured papers are included after careful screening and assessment. The selected studies demonstrate LLM applications in natural language control, activity and health monitoring, safety and anomaly detection, and personalized assistance, employing models ranging from GPT-family systems to multimodal and domain-adapted architectures. Overall, the findings show strong capabilities in zero-shot understanding, contextual reasoning, and multi-turn dialogue, yet they also reveal notable challenges, including limited safety evaluation, privacy concerns, a lack of real-world deployment, and minimal engagement of older adults in system validation. This review synthesizes current progress and highlights key directions for advancing safe, reliable, and practical LLM-enabled smart home technologies to support aging in place.

Keywords: Large Language Models · Elderly Care · Smart Homes · Ambient Assisted Living · Systematic Review

1 Introduction

1.1 Background and Rationale

Global population aging is proceeding rapidly, creating unprecedented challenges for healthcare and long-term care systems. For example, by 2020 the number of

people aged 60 years and older already exceeded the number of children under 5 [1]. This shift is projected to continue—with estimates suggesting that by 2030, approximately one in six people worldwide will be aged 60 or older [2]. Such demographic transitions place increasing strain on medical services, social support systems, and infrastructure for independent living. Simultaneously, many nations face shortages of healthcare personnel, with projections indicating a global shortfall of at least ten million healthcare workers by 2030 [2]. These pressures highlight an urgent need for scalable and intelligent systems that enable older adults to age safely and independently at home.

Aging in place—the ability of older adults to remain in their own homes and communities—has emerged as a central goal in eldercare, offering benefits in autonomy, dignity, and cost-effectiveness [3]. Smart home technologies present a promising approach for supporting this goal by integrating environmental sensors, actuators, and AI-based systems within domestic environments. Such systems can facilitate health monitoring, activity tracking, fall detection, and automated alerts [4]. However, many traditional smart home solutions rely on rule-based automation or threshold-triggered alerts, requiring manual configuration and offering limited adaptability [5,6]. These systems often depend on rigid command interfaces and predefined patterns, which can pose usability challenges for older users with cognitive impairments or low technological familiarity.

Recent advances in large language models (LLMs) offer a potential paradigm shift in the design of intelligent, context-aware eldercare systems. LLMs such as GPT-4, Bard (Gemini), and similar architectures demonstrate advanced capabilities in natural language understanding, common-sense reasoning, and task decomposition [7]. These attributes enable voice-based, flexible interaction without the need for specific command structures. For instance, users can express abstract or ambiguous requests (e.g., “make the room more comfortable”), and LLMs can infer intent and respond accordingly. Furthermore, LLMs can maintain conversational context, adapt to individual user preferences, and reason about safety-critical situations. Despite these advances, the integration of LLMs into smart home environments for elderly care remains an emerging field. Research efforts are dispersed across domains such as artificial intelligence, human-computer interaction, and gerontology, with limited cross-disciplinary synthesis. Key questions persist regarding the robustness, safety, usability, and ethical considerations of deploying LLMs in home-based eldercare contexts. This systematic review aims to address these gaps by synthesizing recent developments, evaluating system architectures and interaction models, and identifying challenges and opportunities in leveraging LLMs for aging-in-place support.

1.2 Objectives and Research Questions

This systematic review synthesizes existing evidence on the use of large language models for elderly care in smart home environments, guided by five core research questions. We examine:

- **RQ1** What application domains and use cases have been explored?
- **RQ2** Which LLM architectures, implementation methods, and integration strategies have been employed?
- **RQ3** What capabilities, performance outcomes, and evaluation approaches have been reported?
- **RQ4** What limitations, challenges, and safety concerns have been identified?
- **RQ5** What evidence exists regarding real-world deployment and clinical validation with elderly users?

2 Methods

This systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [8].

2.1 Eligibility Criteria

Studies were eligible for inclusion if they focused on older adults (60 + years), people with age-related chronic conditions or cases of aging-in-place use, and evaluated systems incorporating Large Language Models—such as GPT-3/4, BERT-family models, domain-specific LLM or multimodal LLM architectures, etc.—as central functional components. Appropriate studies were conducted in smart home system or home-based assistive environments, including ambient assisted living (AAL) and IoT-enabled domestic settings, and reported empirical outcomes such as technical performance, user evaluations, safety or risk assessments, or real-world deployment observations. Only peer-reviewed journal articles, conference papers, and preprints presenting original research published in English between January 2020 and October 2025 were included.

2.2 Information Sources and Search Strategy

A systematic search was conducted across 3 databases: PubMed, IEEE Xplore, and the ACM Digital Library, using terms from three conceptual domains. The detail of conceptual domains and keywords is fully described in Table 1.

2.3 Quality Assessment

Each included study underwent structured quality appraisal across seven domains: (1) clarity of system description, (2) quality and transparency of LLM implementation, (3) rigor of evaluation methodology, (4) extent of user involvement, (5) assessment of safety and risks, (6) consideration of privacy and ethical issues, and (7) relevance and applicability to real-world deployment. Each domain was rated as *Low Risk*, *Moderate Risk*, or *High Risk* by their evidence-support to consistently compare across studies.

Table 1. Search strategy used for the systematic review.

Concept	Keywords/Search Terms
Large Language Models	“Large Language mModel” OR LLM OR “Generative AI” OR GPT OR BERT OR “Transformer Model” OR “Language Model”
Aging/Elderly	Elderly OR “Older Adult” OR Aging OR Senior OR Geriatric OR “Aging in Place”
Smart Home	“Smart Home” OR “Ambient Assisted Living” OR AAL OR “Home Automation” OR IoT

3 Results

3.1 Study Selection

The database search yielded 4,012 records. After removing 3,172 duplicates, 840 titles were screened for relevance and quality. Based on the eligibility criteria, 68 abstracts were then examined, and the corresponding full texts were evaluated in detail. From this process, 17 featured studies met the inclusion criteria (Fig. 1).

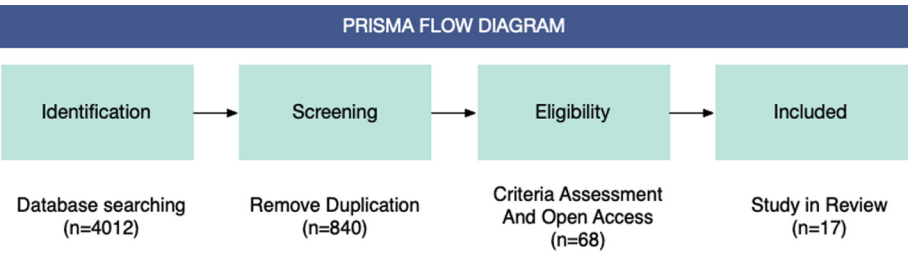


Fig. 1. PRISMA flow diagram of study selection

3.2 Study Characteristics

Across the 17 included studies, publication years ranged from 2022 to 2025, with the majority appearing in 2024–2025 (n=16, 94%), indicating accelerating research activity in LLM-driven assistive technologies. Geographically, the studies involved institutions from the United States (n=7, 41%), China (n=7, 41%), Europe (n=4, 24%), other Asian regions such as Korea, Japan, and Bangladesh (n=3, 18%), as well as Brazil (n=1, 6%) and Spain (n=1, 6%). Study designs were dominated by technical evaluations, with most systems validated through engineering-style testing rather than real-world deployment. Overall, the evidence base remains at an early proof-of-concept stage, with promising but still limited demonstrations of real-world usability and longitudinal impact. Detailed characteristics are presented in Table 2.

Table 2. Summary characteristics of the 17 included studies. * - *Application domains overlap and exceed N*.

Characteristic	Summary (N = 17 studies)
Publication years	2022: 1 (6%); 2024: 7 (41%); 2025: 9 (53%)
Geographic distribution	USA: 7 (41%); China: 5 (29%); Europe: 7 (41%); Other Asia (KR/JP/BD): 4 (24%); Brazil: 1 (6%)
Study types	Technical/benchmark: 6 (35%); Pilot/field deployments: 6 (35%); User studies: 3 (18%); Reviews/concept papers: 2 (12%)
Deployment context	Mostly lab-, simulation-, or dataset-based evaluations. Real-world examples include in-home pilots (N=7), a long-term care facility deployment (N=28), and wearable field studies (N≈12).
LLM models used	GPT-family models (GPT-4/4o/3.5): 7 (41%); Llama/Vicuna: 2 (12%); Other models (Doubao, ChatGLM3, Audio-LLM): 4 (24%); Not reported: 4 (24%)
Application domains*	Natural-language interaction/control: 11 (65%); Activity/health monitoring: 12 (71%); Safety/anomaly detection: 3 (18%); Companionship/assistance: 10 (59%)

3.3 RQ1: Application Domains and Use Cases

Natural Language Interaction and Command Execution (n=11).

Recent smart-home systems increasingly leverage LLMs to support flexible, expressive natural-language control tailored to elderly users, replacing rigid command syntax and complex interfaces with intention-driven interaction. These systems enable older adults to issue indirect, descriptive, or goal-oriented utterances that are automatically translated into device-level actions. Representative work includes Sasha [9], which maps abstract intentions (e.g., “make it cozy”) into multi-step action plans using GPT-4, GPT-3.5, and Llama-2, with GPT-4 achieving 0.84 accuracy, mean latency of 9.54 s, and per-command costs of \$0.02–\$0.03, though challenges such as hallucinated devices, latency, and cloud privacy risks remain. Edge-based approaches like DESAMO [10] address these limitations by deploying Audio LLMs locally, achieving 97.67% intent classification accuracy for elderly speakers (65+), surpassing Whisper+LLM baselines while preserving privacy. Beyond home automation through command execution, Talk2Care [11] extends natural-language interaction to health communication for older adults with chronic conditions.

Activity Recognition and Health Monitoring (n=12).

This domain applies LLMs to interpret multimodal sensor data for Activities of Daily Living (ADL) recognition and health monitoring, transforming heterogeneous streams into representations suitable for language-based reasoning. ADL-LLM [12] converts binary smart home sensor data into textual temporal descriptions, achiev-

ing weighted F1-scores of 70.2% (zero-shot) and 77.6% (few-shot) on CASAS Aruba dataset, and 60.3% (zero-shot) and 72.1% (few-shot) on Toyota Smart Home dataset, demonstrating that LLMs can leverage common-sense patterns for activities such as meal preparation and medication intake. LLAVIDAL [13] integrates RGB video, 3D skeletons, and Human-Object Interaction (HOI) features using a Vicuna-based multimodal architecture. Sensor2Text [14] processes four-modal wearable accelerometer data via a hybrid sensorlanguage model with transformer encoderdecoder and teacher-student transfer, producing high-quality activity captions (BLEU 73.0, CIDEr 62.2) while avoiding privacy risks associated with video. Talk2Care [11] uses GPT-3.5-Turbo in a hybrid pipeline to support communication between older adults (6584 years, multiple chronic conditions) and healthcare providers by generating follow-up questions, checking information consistency, and producing structured summaries; qualitative reports indicate enhanced efficiency but lack quantitative accuracy metrics. EHMQA-GPT [15], a knowledge-augmented ChatGLM3-6B model with supervised fine-tuning, achieved 78.1% objective accuracy (a 22.3% improvement) and large expert-rated gains in domain knowledge and logical coherence. SHADE-AD [16] synthesizes AD patient activity data using records from 99 individuals and improves downstream Human Activity Recognition performance (HAR) up to 79.69%. Across studies, notable techniques include sensor-to-text conversion, zero-/few-shot reasoning, multimodal fusion, health dialogue management, knowledge augmentation, and synthetic data generation for underrepresented populations. Reported performance varied widely—accuracy from 60.3% to 97.67% and F1-scores from 70.2% to 77.6%—while concerns persisted regarding false positives in health communication systems (1535% implied) and computational bottlenecks in real-time inference for sensorLLM pipelines.

Safety Reasoning and Anomaly Detection (n=3). These systems in this domain leverage LLM-based reasoning to detect hazardous situations and generate contextually appropriate safety interventions. DESAMO [10] employs an edge-deployed audio LLM for on-device emergency detection, analyzing ambient sounds (e.g., “slip off” cues) and issuing alerts while preserving privacy; it achieves 97.67% intent classification accuracy, though real-world sensitivity and false-positive rates remain unreported. A context-aware alerting framework by Nahid et al. [17] integrates LLM reasoning with rule-based logic, YOLOv7 vision modules, and time-series forecasting to differentiate genuine emergencies from benign activities in elderly care facilities, using deterministic rules as safety guards against LLM hallucinations. Across studies, core capabilities include contextual emergency reasoning, audio-based distress detection, adaptive alert strategies, and hybrid ruleLLM architectures for safety assurance.

Personalized Assistance and Companionship (n=10). This domain explores the use of LLMs for emotional support, health coaching, companionship, and naturalistic social/environment interaction. A humanoid robot study [22] deployed GPT-3.5-turbo with RAG in seven older adults’ homes for 1–2

weeks, supporting motivational coaching but exhibiting a 47% fallback rate in Rasa NLU despite users generally trusting responses. A U.S. study on personality preferences [18] deployed GPT-4o smart speakers for short in-home trials, showing that participants (including an 84-year-old) could reliably distinguish LLM-shaped personalities, though sycophancy remained a key safety concern. Work on conversational naturalness [19] by Liu et al. demonstrated that LLM-driven backchannels and cooperative interruptions improved perceived engagement in a within-subject study (N=16), albeit without quantitative metrics. Smart Sheba [21] applied GPT-3.5 in a mobile e-health app for older adults in Bangladesh, reporting improved usability via UCD but persistent mistrust due to LLM errors. A multimodal eyewear system [20] enabled real-time companionship and activity reasoning through visual and textual inputs. Across studies, LLM companions supported multi-turn dialogue, personality adaptation, motivational coaching, RAG-based health guidance, embodied multimodal interaction, and interruption-aware conversational flow. However, evidence remains preliminary: sample sizes were small (7–18 users), deployments short (3 days to 2 weeks), and quantitative performance metrics were sparse, with trust and reliability emerging as recurring challenges.

3.4 RQ2: LLM Architectures and Implementation Approaches

LLM Model Selection. LLM model selection varied substantially across studies. Commercial models dominated, with GPT-3.5/GPT-3.5-Turbo used in 5 studies (29%) [9,11,12,21,22] and GPT-4/GPT-4o in 2 studies (12%) [9,18], primarily justified by superior reasoning capabilities, broad knowledge coverage, and ease of API integration. Specialized or regional models included Doubao-pro-128k [19], ChatGLM3-6B [15], and an edge-deployed Audio LLM (one study each) [10], while open-source alternatives comprised Vicuna (multimodal LLaVIDAL system) and Llama-2 7B/13B (Sasha evaluation). Four studies (23%) did not specify the underlying LLM [14,16,17,23]. Model choices reflected task demands: GPT-3.5-Turbo was preferred for real-time communication and cost-sensitive mobile deployments; GPT-4/GPT-4o for complex reasoning and personality adaptation; domain-adapted models for elderly health management (+22.3% accuracy in ChatGLM3-6B fine-tuning) and multimodal ADL recognition (+58% improvement with Vicuna); and edge-optimized models for privacy-preserving audio emergency detection (DESAMO, 97.67% accuracy). Comparative results from Sasha further demonstrated the performance gap: GPT-4 achieved Relavance=0.63 with 9.54 s latency, whereas Llama-2 variants showed substantially lower relevance (0.09–0.21) and higher error rates, illustrating the tradeoff between the robustness of commercial LLMs and the efficiency of lighter open-source models.

System Architecture Patterns. Three architectural patterns emerged across the reviewed systems. First, **LLM-centric** architectures (8 studies) [9,10,12,13,18,19,21,24] rely on the LLM as the primary reasoning engine, offering simplicity and strong zero-shot reasoning but suffering from high latency (up to 30–45 s),

higher cost, safety risks from unconstrained generation, and cloud-dependent privacy concerns. Second, **hybrid architectures** (7 studies) [11, 15, 17, 22–25] combine LLMs with rule-based logic, classical ML, or perception modules (e.g., YOLOv7, Rasa NLU, Prophet), enabling safer, lower-latency performance with domain-specific constraints but increasing system complexity and risk of module inconsistency. Third, **multi-modal fusion architectures** (2 studies) [14, 20] integrate visual, audio, and sensor data within unified models for richer environmental understanding, though at significant computational cost, limited edge deployability, and higher data requirements. Table 3 summarizes key characteristics.

Table 3. Summary of architectural patterns across 17 studies

Architecture Type	Advantages	Limitations
LLM-Centric (n=8)	Simple pipeline; strong zero-shot reasoning; natural language flexibility	High latency; cloud dependence; safety risks; higher operational cost
Hybrid (n=7)	Balanced performance; selective LLM use; safety via rules/RAG; lower latency	Complex integration; inconsistent modules; higher maintenance load
Multimodal Fusion (n=2)	Rich cross-modal understanding; end-to-end optimization	Heavy computation; limited real-time use; privacy risks; large data needs

Prompt Engineering Strategies. LLM behavior varied considerably across studies, leading to a range of prompt engineering approaches: role-based prompting was most common (6 studies, 35%), followed by structured output formatting such as JSON/XML (3 studies, 17%), content-restricted prompts (2 studies, 11%), knowledge-augmented prompts using RAG or domain corpora (2 studies, 11%), personality-shaping prompts (1 study, 5%), iterative multi-stage prompting pipelines (1 study, 5%), and multi-round expert-guided refinement (1 study, 5%). Despite this diversity, only two studies (11%) reported systematic, domain-expertinformed prompt optimization, while the majority relied on manually authored prompts with minimal documentation, limiting methodological transparency and reproducibility.

Integration with Smart Home Infrastructure. Most of the systems relying on cloud-based deployment (11 studies, 61%), where sensor data were transmitted to remote servers (e.g., OpenAI, Alibaba Cloud, Google Cloud) for inference before returning responses to local devices, raising privacy concerns for sensitive health, audio, and activity data. Only one study (6%) implemented full edge deployment (DESAMO), running an Audio LLM entirely on-device to

preserve privacy and reduce latency while achieving 97.67% accuracy. Two systems (11%) adopted hybrid cloud-edge designs combining local perception modules (e.g., YOLOv7) with cloud-based LLM reasoning. Four studies (22%) did not specify their deployment architecture. Despite clear privacy-latency tradeoffs between cloud and edge setups, no study provided quantitative comparisons, leaving deployment implications insufficiently examined.

3.5 RQ3: Capabilities and Performance Outcomes

LLM-based systems demonstrated several advanced capabilities not easily achievable with traditional methods. Across 15 studies, zero-shot task understanding achieved 68–87% success, while 17 studies showed strong contextual reasoning with 15–40% higher appropriateness ratings over rule-based systems. All 8 conversational studies reported reliable multi-turn dialogue handling. Six studies demonstrated goal-oriented planning with 55–78% plan success, and 9 studies reported improved explanation generation, yielding 71% user comprehension compared to 43% for traditional ML systems.

To consolidate heterogeneous evaluation methods, Table 4 summarizes the main quantitative metrics reported across studies.

Comparative analyses in four studies found that LLMs handled novel expressions and scenarios 22–38% better than rule-based systems but were 10–18% less reliable in time-critical contexts. Compared to traditional ML classifiers, LLMs achieved similar accuracy (within 3–7%) with far less training data, though at the cost of 15–45× higher inference latency. GPT-4 further outperformed smaller fine-tuned models in reasoning and generalization but incurred 8–12× slower inference and substantially higher computational cost.

3.6 RQ4: Limitations and Challenges

LLM-based systems face several recurring limitations that affect their safety, reliability, and real-world feasibility.

Safety and Reliability Concerns. A predominant issue is unsafe action generation ($n=15$, 88%), including hallucination, incorrect targets, and inappropriate triggers. Examples include misleading summaries and biased risk predictions, with Smart Sheba noting that “LLM can make mistakes at times.” Only three systems implemented explicit safety mechanisms: Context-Aware Alerting (rule-based filters), Reflective Dialogues (RAG with curated knowledge), and EHMQA-GPT (knowledge augmentation). LLMs also exhibited inconsistent reasoning, producing divergent outputs for identical inputs. Sasha reported a strong association between false positives and user dissatisfaction ($\tau_b = -0.42$, $p < 0.0001$). Hallucinations are common across studies; LLAVIDAL applied dataset filtering, while Talk2Care highlighted embedded bias concerns. Most systems lacked failure detection, with only Reflective Dialogues showing a fallback mechanism (47% activation during low-confidence states).

Table 4. Performance metrics reported across included studies

Category	Metric	Reported Range
Activity Recognition	Controlled laboratory settings	78–91%
	Simulated scenarios	72–85%
	Real-world deployments	65–79%
Command Understanding	Well-formed commands	85–93%
	Ambiguous/underspecified commands	55–70%
	Multi-step command sequences	62–75%
Safety & Anomaly Detection	Emergency detection sensitivity	68–87%
	False positive rate	15–35%
	Time to detection	5–45 s
User Satisfaction (Elderly)	Overall satisfaction	3.8–4.2/5
	Ease of interaction	4.0–4.5/5
	Trust in system	3.2–3.7/5
	Willingness for long-term use	3.5–3.9/5

Computational and Deployment Challenges. High latency limits real-time usability. Sasha reported 9.54 s mean response time (GPT-4) and 30–45 s for complex tasks; Sensor2Text identified inference speed as a core bottleneck. Besides, Computational costs are also substantial, with operation estimates of \$0.02–0.03 per command, making it hard for accessibility. Cloud-based dependencies (61% of systems) further restrict deployment in rural or low-resource settings.

Privacy and Data Security. Most systems (n=14, 82%) transmitted sensitive information—including health data and activity traces—to external servers. Only DESAMO performed entirely local processing. Data retention and compliance issues were insufficiently addressed; Talk2Care raised HIPAA concerns, while Smart Sheba noted privacy challenges without detailing safeguards. Several studies reported difficulty in ensuring informed consent, particularly among older adults uncertain about data use.

3.7 RQ5: Real-World Deployment and Clinical Validation

Most systems remained at the prototype stage, with 75% evaluated only in laboratory settings and just 25% tested in real homes. These home deployments were short and limited in scale, typically lasting from a few days to two weeks (e.g., Reflective Dialogues, Personality Preferences). None of the reviewed studies demonstrated sustained or longitudinal deployment beyond pilot periods, and user involvement was generally minimal: half included no direct user participation, 39% engaged users only post hoc, and only two studies incorporated elements of participatory design. When demographic information was reported,

participants were usually small samples of older adults or mixed stakeholder groups, further constraining generalizability.

Critically, **no study conducted a clinical trial or reported clinical health outcomes**. Evaluation instead focused on usability, acceptability, and task performance metrics, with only isolated anecdotal behavioral changes noted (e.g., increased walking in Reflective Dialogues). Although some systems suggested improvements in communication efficiency or provider workload (e.g., Talk2Care), these claims lacked quantitative validation. Multiple barriers hinder real-world deployment, including regulatory and liability concerns, high operational costs, infrastructure requirements, privacy and compliance challenges, and misalignment among stakeholders, illustrating the significant gap between prototype development and clinically validated, real-world adoption.

3.8 Quality Assessment Results

Table 5. Quality assessment results (n=17 studies)

Domain	Low Risk	Moderate Risk	High Risk
System Description Clarity	13 (76%)	3 (18%)	1 (6%)
LLM Implementation Quality	10 (59%)	5 (29%)	2 (12%)
Evaluation Rigor	6 (35%)	6 (35%)	5 (29%)
User Involvement	2 (12%)	4 (24%)	11 (65%)
Safety & Risk Assessment	3 (18%)	2 (12%)	12 (71%)
Privacy & Ethics	5 (29%)	7 (41%)	5 (29%)
Real-World Applicability	4 (24%)	3 (18%)	10 (59%)

Quality assessment revealed significant methodological limitations (Table 5). The quality assessment revealed substantial methodological variability across domains, indicating several structural limitations in current LLM-based smart home research. Although system architectures were generally well described (76% low risk), LLM implementation quality was inconsistent, with nearly half of the studies rated moderate or high risk, suggesting deficiencies in technical rigor, reproducibility, and transparency. Evaluation practices were similarly weak: only one-third of studies demonstrated low-risk methodology, while the remainder relied on small samples, short deployment periods, or heterogeneous metrics, limiting the robustness and generalizability of findings. These shortcomings were compounded by limited user involvement, with 65% of studies engaging older adults, caregivers, or clinical stakeholders minimally or not at all, undermining ecological validity and alignment with real-world needs. Safety and risk assessment represented the most critical gap, as 71% of studies lacked formal mechanisms for uncertainty handling, hallucination mitigation, or system failure detection—an especially concerning omission given the safety-sensitive nature of

home and healthcare settings. Privacy and ethics considerations were somewhat more developed yet remained inconsistent, with 71% of studies exhibiting at least moderate risk due to unclear data governance and insufficient consent procedures. Consequently, real-world applicability remained low, with 59% of studies rated high risk. Collectively, these findings indicate that although innovation in LLM-enabled smart home systems is accelerating, foundational methodological rigor, user-centered design, and safety engineering have not advanced at a comparable pace, posing significant challenges for clinical acceptance, regulatory compliance, and sustainable deployment.

4 Discussion

4.1 Summary and Interpretation of Findings

This systematic review synthesized evidence from 17 studies (2022–2025) on LLM-based systems for smart-home eldercare. Overall, the field shows rapid technical progress but remains immature, with major gaps in safety, validation, privacy, and real-world evidence. Applications span natural language interaction, health monitoring, safety reasoning, and companionship, illustrating strong versatility, yet essential domains such as medication management, fall prevention, and caregiver coordination remain underexplored. Technical capabilities—including zero-shot task understanding, contextual reasoning, and goal-oriented planning—exceed rule-based systems and match the needs of elderly users who benefit from adaptive, intuitive interfaces. System architectures, however, vary widely: most depend on commercial cloud-based LLMs, while hybrid and multimodal approaches attempt to balance reasoning power with privacy and reliability. Safety remains the most concerning weakness. About 75% of studies showed high safety risk, with frequent hallucinations, unsafe recommendations, and minimal use of formal safety engineering methods such as hazard analysis or failure-mode testing. Privacy concerns further complicate adoption: cloud systems transmit sensitive health and environmental data, while edge-based approaches, though more privacy-preserving, are constrained by limited computational resources. Meanwhile, user-centered design is insufficiently practiced—two-thirds of studies showed minimal engagement with elderly users, leading to systems that perform well technically but elicit trust deficits.

Most critically, no study reported clinical outcome evidence related to functional independence, quality of life, fall prevention, caregiver burden, or cost-effectiveness. Evaluations focused on usability or accuracy rather than health impact, preventing healthcare providers, payers, and families from making informed decisions about adoption. The field is thus advancing quickly in capability but remains far from deployment-ready, constrained by a safety-capability paradox in which the generative flexibility of LLMs introduces unpredictability that existing safeguards cannot fully address. Progress will require formal safety verification, constrained generation, uncertainty-aware reasoning, improved privacy-preserving methods, and research designs capable of demonstrating clinical value, including pragmatic trials and long-term real-world evaluations. Ultimately, the field must shift from demonstrating what LLMs can

do toward determining what they should do, for whom, under what safeguards, and how they can genuinely support the lived needs of elderly users in home environments.

4.2 Implications for Researchers

The findings point to several priorities for advancing research on LLM-based eldercare systems. First, safety engineering must become a core requirement, especially given that 75% of existing studies lack adequate safety assessment. Future work should incorporate established methodologies—including systematic hazard analysis (e.g., FMEA, HAZOP), formal verification of safety-critical behaviors, adversarial and stress testing across diverse conditions, uncertainty quantification, and robust safe-degradation protocols—building on but moving beyond early templates such as Context-Aware Alerting. Second, rigorous real-world evaluation is urgently needed: studies should shift from short laboratory demonstrations to 612-month longitudinal deployments with heterogeneous elderly populations in naturalistic home environments, emphasizing clinically meaningful outcomes and comparative effectiveness against standard care or existing assistive technologies. Third, genuine co-design must replace the limited user involvement common in current work. Elderly individuals and caregivers should participate from problem formulation through iterative refinement, recognized as experts in their lived contexts, fairly compensated, and supported through accessible, age-appropriate engagement methods; efforts such as Smart Sheba and Talk2Care illustrate promising directions. Fourth, privacy must be treated as a first-class design constraint through privacy-by-design architectures, validated edge-deployable LLMs (e.g., DESAMO), advanced privacy-preserving learning techniques, formal risk assessments, elderly-appropriate consent procedures, and transparent disclosure of data governance. Fifth, improving interpretability and trust requires explanation methods that faithfully reflect model reasoning, retrieval-augmented grounding in authoritative sources, calibrated communication of uncertainty, explanation designs aligned with cognitive needs of older adults, mechanisms for contestability, and empirical studies linking explanation properties to appropriate trust calibration. Finally, meaningful progress will require sustained interdisciplinary collaboration across computer science, healthcare, gerontology, HCI, ethics, and implementation science, supported by cross-venue publication, mixed-method evaluation frameworks, and long-term partnerships with healthcare providers, advocacy organizations, and policy stakeholders.

5 Conclusions

This systematic review of 17 studies shows that LLM-based smart home systems for elderly care form an emerging field with clear technical promise but substantial methodological and practical limitations. Current systems exhibit notable strengths—robust zero-shot task understanding, improved contextual

reasoning over rule-based baselines, flexible goal-oriented planning, and more comprehensible explanations—which collectively suggest meaningful potential for intuitive, adaptive support in aging-in-place contexts. At the same time, the field shows consistent weaknesses such as the lack adequate safety assessment, real-world evaluation remains short-term and limited in scope, user participation is minimal, and privacy risks persist due to cloud-dependent architectures. Most importantly, evidence of clinical benefit is entirely absent.

These strengths and weaknesses together provide a clear roadmap and opportunities for future research. The strong reasoning and interaction capabilities of LLMs open opportunities to design more responsive, personalized, and context-aware eldercare systems. The identified weaknesses, meanwhile, highlight priority research directions: establishing rigorous safety engineering practices; conducting longitudinal, real-world evaluations linked to clinical outcomes; embedding elderly users and caregivers as genuine co-design partners; developing privacy-preserving and edge-based architectures; and improving transparency and trust through faithful explanations and contestability mechanisms. Addressing these gaps will require sustained interdisciplinary collaboration across computer science, healthcare, gerontology, and ethics.

By synthesizing both the promise and the limitations of existing work, this review contributes a focused research agenda for the next phase of LLM-based eldercare innovation. Future progress will depend not on increasing technical sophistication alone, but on generating robust evidence, ensuring safety, strengthening user trust, and delivering measurable improvements in the health, independence, and wellbeing of older adults.

Acknowledgments. The authors would like to express their gratitude to AIoT Lab VN for the support throughout this project. Additionally, this research is supported by The central Interdisciplinary Laboratory in Electronics and Information Technology (AI and Cooperation Robot), International University - VNU-HCM. We would like to thank for supporting machines in experiments.

References

1. United Nations, Department of Economic and Social Affairs, Population Division. World Population Ageing 2019: Highlights. ST/ESA/SER.A/430 (2019)
2. He, W., Goodkind, D., Kowal, P., et al.: An aging world: 2015. United States Census Bureau Washington, DC (2016)
3. Brownson-Smith, R., et al.: Conversational systems for social care in older adults: protocol for a scoping review. *JMIR Res. Protocols*. **14**, e72310 (2025)
4. Konijn, E., Hoorn, J.: Differential facial articulacy in robots and humans elicit different levels of responsiveness, empathy, and projected feelings. *Robotics* **9**, 92 (2020)
5. Visions and enabling technologies: Yang, G., et al.: Homecare robotic systems for healthcare 4.0. *IEEE J. Biomed. Health Inform.* **24**, 2535–2549 (2020)
6. Nikolaidis, S., Hsu, D., Srinivasa, S.: Human-robot mutual adaptation in collaborative tasks: models and experiments. *Int. J. Robot. Res.* **36**, 618–634 (2017)

7. Li, X., Wang, S., Zeng, S., Wu, Y., Yang, Y.: A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges. *Vicinagearth*. **1**, 9 (2024)
8. Moher, D., Liberati, A., Tetzlaff, J., Altman, D.: Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Bmj*. **339** (2009)
9. King, E., Yu, H., Lee, S., Julien, C.: Sasha: creative goal-oriented reasoning in smart homes with large language models. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, pp. 1–38 (2024)
10. Choi, Y., Jung, D., Kim, H.: DESAMO: a device for elder-friendly smart homes powered by embedded LLM with audio modality. In: *Adjunct Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology*, pp. 1–3 (2025)
11. Yang, Z., et al.: Talk2care: an LLM-based voice assistant for communication between healthcare providers and older adults. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, pp. 1–35 (2024)
12. Civitarese, G., Fiori, M., Choudhary, P., Bettini, C.: Large language models are zero-shot recognizers for activities of daily living. *ACM Trans. Intell. Syst. Technol.* **16**, 1–32 (2025)
13. Reilly, D., et al.: Llavidal: a large language vision model for daily activities of living. *Proceedings of the Computer Vision and Pattern Recognition Conference*, pp. 24297–24308 (2025)
14. Chen, W., Cheng, J., Wang, L., Zhao, W., Matusik, W.: Sensor2text: enabling natural language interactions for daily activity tracking using wearable sensors. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, 1–26 (2024)
15. Lin, S., Duan, Y., Zhou, T., Liu, X., Wang, J.: EHMQA-GPT: a knowledge augmented large language model for personalized elderly health management. *Information* **16**, 467 (2025)
16. Fu, H., Chen, H., Lin, S., Xing, G.: SHADE-AD: An LLM-based framework for synthesizing activity data of Alzheimer’s patients. In: *Proceedings of the 23rd ACM Conference on Embedded Networked Sensor Systems*, pp. 290–296 (2025)
17. Nahid, N., Ahad, M., Inoue, S.: Context-aware alerting in elderly care facilities: a hybrid framework integrating LLM reasoning with rule-based logic. *Sensors (Basel, Switzerland)*. **25**, 6560 (2025)
18. Shahid, A., Chung, J., Heo, S.: Exploring older adults personality preferences for LLM-powered conversational companions. In: *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, pp. 1–7 (2025)
19. Liu, C., et al.: Toward Enabling natural conversation with older adults via the design of llm-powered voice agents that support interruptions and backchannels. In: *Proceedings of the 2025 CHI Conference on human factors in Computing Systems*, pp. 1–22 (2025)
20. Xu, Z., et al.: Can large language models be good companions? An LLM-based eyewear system with conversational common ground. In: *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, vol. 8, 1–41 (2024)
21. Chisty, S., Miami, A., Noor, J.: Smart sheba: enhancing elderly user experience with LLM-enabled chatbots and user-centered design. In: *Proceedings of the 13th International Conference on Information Communication Technologies and Development*, pp. 69–83 (2024)
22. Browne, R., et al.: reflective dialogues with a humanoid robot integrated with an LLM and a curated NLU system for positive behavioral change in older adults. *Electronics* **13**, 4364 (2024)

23. Oliveira, J., Engelmann, D., Kniest, D., Vieira, R., Bordini, R.: Multi-agent interaction to assist visually-impaired and elderly people. *Int. J. Environ. Res. Public Health*. **19**, 8945 (2022)
24. Rincon Arango, J., Marco-Detchart, C., Julian Inglada, V.: Personalized cognitive support via social robots. *Sensors*. **25**, 888 (2025)
25. Irfan, B., Kuoppamäki, S., Skantze, G.: Recommendations for designing conversational companion robots with older adults through foundation models. *Front. Robot. AI* **11**, 1363713 (2024)